Alien Dictionary (hard)

Problem Statement

There is a dictionary containing words from an alien language for which we don’t know the ordering of the letters. Write a method to find the correct order of the letters in the alien language. It is given that the input is a valid dictionary and there exists an ordering among its letters.

Example 1:

```
Input: Words: ["ba", "bc", "ac", "cab"]
Output: bac
Explanation: Given that the words are sorted lexicographically by the
rules of the alien language, so
from the given words we can conclude the following ordering among its
characters:

1. From "ba" and "bc", we can conclude that 'a' comes before 'c'.
2. From "bc" and "ac", we can conclude that 'b' comes before 'a'

From the above two points, we can conclude that the correct character
order is: "bac"
```

Example 2:

```
Input: Words: ["cab", "aaa", "aab"]
Output: cab
Explanation: From the given words we can conclude the following ordering
among its characters:

1. From "cab" and "aaa", we can conclude that 'c' comes before 'a'.
2. From "aaa" and "aab", we can conclude that 'a' comes before 'b'

From the above two points, we can conclude that the correct character
order is: "cab"
```

Example 3:

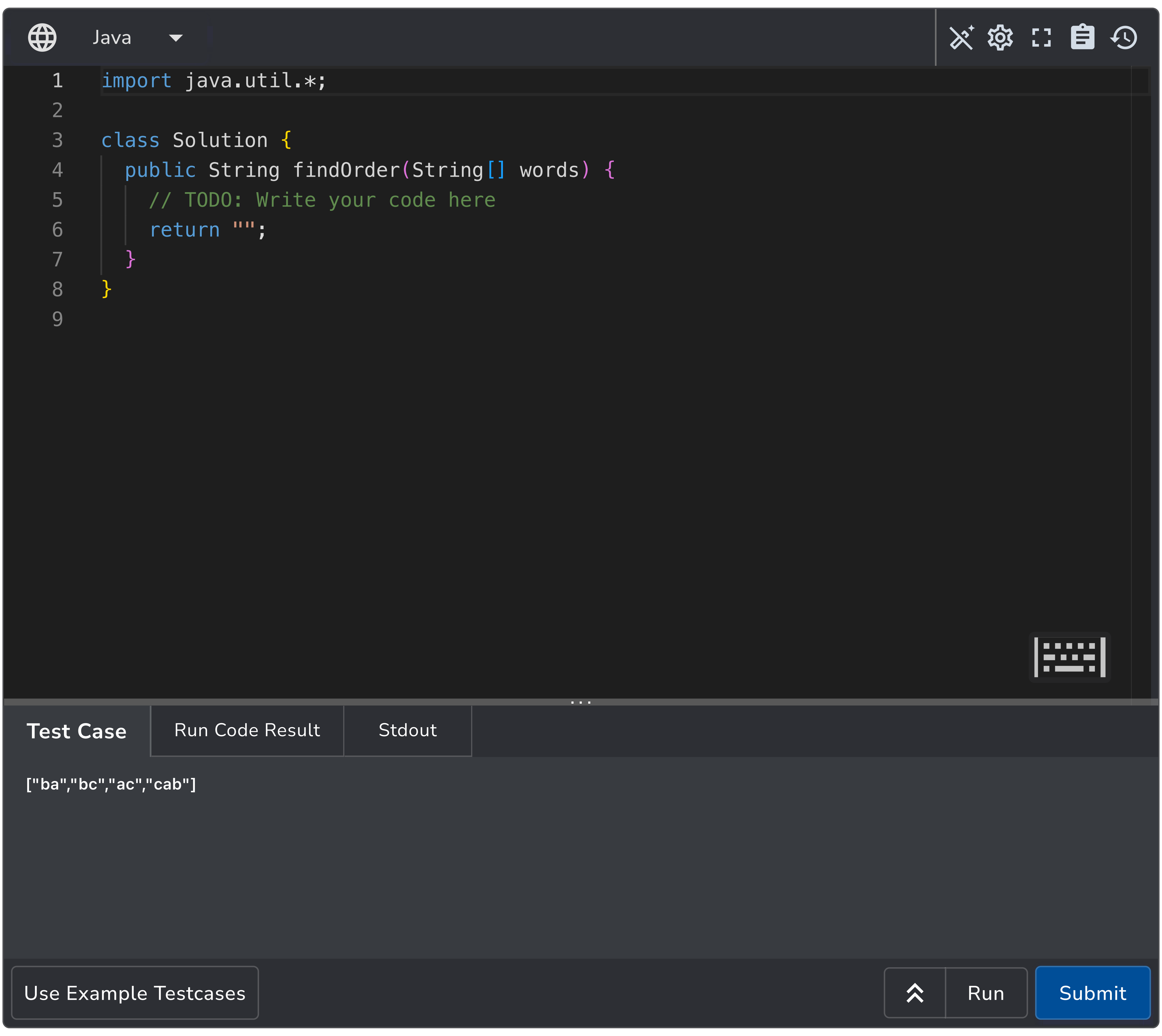
```
Input: Words: ["ywx", "wz", "xww", "xz", "zyy", "zwz"]
Output: ywxz
Explanation: From the given words we can conclude the following ordering
among its characters:

1. From "ywx" and "wz", we can conclude that 'y' comes before 'w'.
2. From "wz" and "xww", we can conclude that 'w' comes before 'x'.
3. From "xww" and "xz", we can conclude that 'w' comes before 'z'
4. From "xz" and "zyy", we can conclude that 'x' comes before 'z'
5. From "zyy" and "zwz", we can conclude that 'y' comes before 'w'

From the above five points, we can conclude that the correct character
order is: "ywxz"
```

Try it yourself

Try solving this question here:



```java
import java.util.*;

class Solution {
  public String findOrder(String[] words) {
    // TODO: Write your code here
    return "";
  }
}
```

```
["ba","bc","ac","cab"]
```